

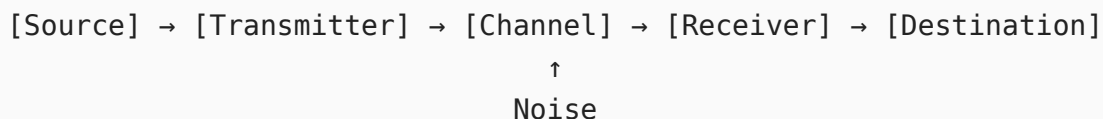
# Principles of Communications - CIA 1 Answer Key

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## Set A

### Part-A

#### 1. Draw the block diagram of communication systems.



#### 2. Compare DSB-SC and SSB.

| Parameter   | DSB-SC     | SSB      |
|-------------|------------|----------|
| Sidebands   | Both       | One      |
| Bandwidth   | $2f_m$     | $f_m$    |
| Complexity  | Moderate   | High     |
| Application | Data links | HF voice |

#### 3. For an FM system, the maximum deviation is 75 kHz and the highest modulating frequency is 15 kHz. Determine the required transmission bandwidth using Carson's rule.

$$B_T = 2(\Delta f + f_m) = 2(75 + 15) = \boxed{180 \text{ kHz}}$$

#### 4. Identify suitable modulation technique for TV transmission and justify.

**VSF (Vestigial Sideband)** - transmits full USB + a vestige of LSB, fitting within the 6 MHz channel allocation while avoiding the need for an ideal sharp-cutoff filter that SSB would demand near DC.

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## 5. Classify the FM signal as NBFM or WBFM if $\beta = 0.5$ .

**NBFM** - since  $\beta = 0.5 < 1$ , only carrier and two sidebands are significant; bandwidth  $\approx 2f_m$ .

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## 6. Summarize the key differences between a random variable and a random process.

- A **random variable** maps one experiment outcome to a single number
  - A **random process**  $\{X(t)\}$  maps each outcome to an entire time waveform - it's an indexed family of random variables
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## 7. Discuss the mathematical conditions for Wide Sense Stationary.

A process  $X(t)$  is WSS if:

$$E[X(t)] = \mu \quad (\text{constant}) \quad \text{and} \quad R_X(t_1, t_2) = R_X(\tau), \quad \tau = t_1 - t_2$$

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## 8. Show how narrowband noise behaves in a modulated system and state its characteristics.

Narrowband noise is written as  $n(t) = n_I(t) \cos \omega_c t - n_Q(t) \sin \omega_c t$ , where:

- $n_I(t)$ ,  $n_Q(t)$  are lowpass, equal power, and mutually uncorrelated
  - In AM detection only  $n_I$  appears at output; in FM,  $n_Q$  dominates with a parabolic ( $f^2$ ) output PSD
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## 9. Apply the noise equivalent bandwidth formula to analyze receiver performance.

$$B_{eq} = \frac{\int_0^\infty |H(f)|^2 df}{|H_{max}|^2}, \quad N_o = N_0 |H_{max}|^2 B_{eq}$$

$B_{eq}$  is always wider than the 3 dB bandwidth (e.g.,  $B_{eq} = 1.57 f_3$  for a single-pole RC filter), so actual noise power is higher than the 3 dB bandwidth alone would suggest.

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## Part-B

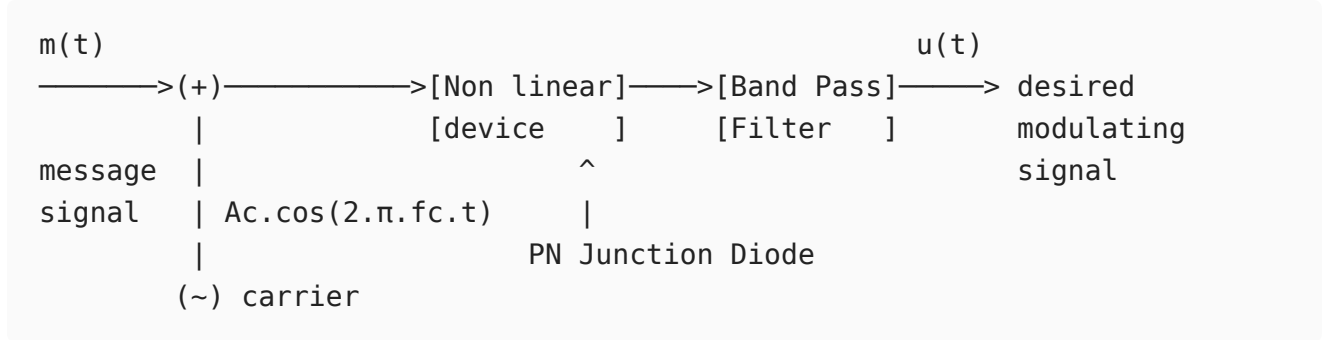
**i. Demonstrate how an AM signal is generated using a square law modulator and derive its mathematical equation.**

**Principle:**

A square-law modulator uses a nonlinear device (diode or FET biased in the square-law region) whose transfer characteristic approximates:

$$v_o = a_1 v_i + a_2 v_i^2$$

**Block Diagram:**



**Derivation:**

The input to the device:

$$v_i(t) = A_c \cos \omega_c t + m(t)$$

Applying the square-law characteristic:

$$v_o = a_1 [A_c \cos \omega_c t + m(t)] + a_2 [A_c \cos \omega_c t + m(t)]^2$$

Expanding the squared term:

$$v_o = a_1 A_c \cos \omega_c t + a_1 m(t) + a_2 A_c^2 \cos^2 \omega_c t + 2a_2 A_c m(t) \cos \omega_c t + a_2 m^2(t)$$

Using  $\cos^2 \omega_c t = \frac{1}{2} (1 + \cos 2\omega_c t)$ :

$$v_o = \underbrace{a_1 m(t) + a_2 m^2(t)}_{\text{lowpass terms}} + \underbrace{\left[ a_1 A_c + 2a_2 A_c m(t) \right] \cos \omega_c t}_{\text{AM term at } f_c} + \underbrace{\frac{a_2 A_c^2}{2} \cos 2\omega_c t}_{\text{at } 2f_c}$$

After the BPF centered at  $f_c$  passes only the term at  $\omega_c$ :

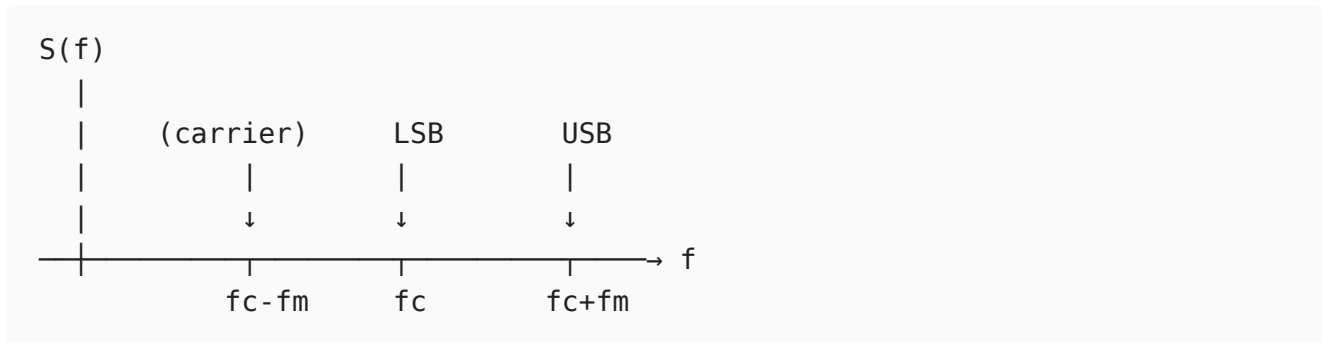
$$s_{AM}(t) = a_1 A_c \left[ 1 + \frac{2a_2}{a_1} m(t) \right] \cos \omega_c t$$

This is the standard AM signal with modulation sensitivity  $k_a = \frac{2a_2}{a_1}$ .

For a single-tone message  $m(t) = A_m \cos \omega_m t$ , the modulation index is:

$$\mu = k_a A_m = \frac{2a_2 A_m}{a_1}$$

## Spectrum of the output:



The distortion term  $a_2 m^2(t)$  is removed by the BPF. The output power is:

$$P_{AM} = \frac{(a_1 A_c)^2}{2} \left( 1 + \frac{\mu^2}{2} \right)$$

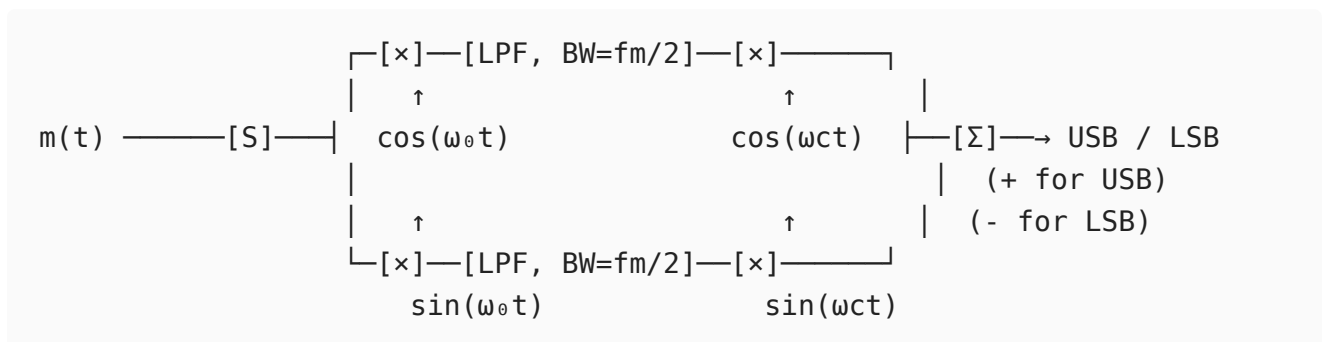
**Limitation:** The square-law approximation is valid only for small signal swings. For larger signals, higher-order nonlinear terms introduce additional harmonic distortion components that must be filtered out.

## ii. Construct the block diagram based on Weaver's method and describe how it generates an SSB signal.

### Weaver's Method (Third Method):

Weaver's method generates SSB using two stages of modulation and lowpass filters, avoiding the need for an ideal sharp-cutoff sideband filter.

### Block Diagram:



where  $f_0$  = midband frequency of message (typically  $\frac{f_{m,max} + f_{m,min}}{2}$ , e.g., 2 kHz for audio 300 Hz - 3.4 kHz).

**Step-by-step mathematical analysis** for  $m(t) = A_m \cos \omega_m t$ :

### Upper path:

After first multiplier:

$$u_1(t) = A_m \cos \omega_m t \cdot \cos \omega_0 t = \frac{A_m}{2} [\cos(\omega_m - \omega_0)t + \cos(\omega_m + \omega_0)t]$$

After LPF (retains  $\omega_m - \omega_0$  component, since  $|\omega_m - \omega_0| < \omega_0$ ):

$$u_2(t) = \frac{A_m}{2} \cos(\omega_m - \omega_0)t$$

After second multiplier with  $\cos \omega_c t$ :

$$u_3(t) = \frac{A_m}{4} [\cos(\omega_c + \omega_m - \omega_0)t + \cos(\omega_c - \omega_m + \omega_0)t]$$

**Lower path:**

After first multiplier with  $\sin \omega_0 t$ :

$$l_1(t) = \frac{A_m}{2} [\sin(\omega_m - \omega_0)t + \sin(\omega_m + \omega_0)t]$$

After LPF:

$$l_2(t) = \frac{A_m}{2} \sin(\omega_m - \omega_0)t$$

After second multiplier with  $\sin \omega_c t$ :

$$l_3(t) = \frac{A_m}{4} [\cos(\omega_c - \omega_m + \omega_0)t - \cos(\omega_c + \omega_m - \omega_0)t]$$

**Adding (USB):**

$$s_{USB}(t) = u_3 + l_3 = \frac{A_m}{2} \cos(\omega_c + \omega_m - \omega_0)t$$

**Subtracting (LSB):**

$$s_{LSB}(t) = u_3 - l_3 = \frac{A_m}{2} \cos(\omega_c - \omega_m + \omega_0)t$$

**Advantages:**

- Avoids sharp-cutoff filters - LPFs are easy to build
- Requires only a 90° phase shift at fixed frequency  $f_0$ , not across the whole signal band
- Well-suited for digital/DSP implementation

**Disadvantage:** Any imbalance in the two paths introduces the unwanted sideband (carrier/sideband suppression depends on component matching). The DC offset at the second LPF output also creates a tone at  $f_c - f_0$  if not carefully handled.

i. Apply angle modulation principles to formulate the mathematical expressions of FM and PM waves.

**General Angle Modulated Signal:**

$$s(t) = A_c \cos[\omega_c t + \phi(t)]$$

where  $\phi(t)$  is the instantaneous phase deviation. The instantaneous frequency is:

$$f_i(t) = f_c + \frac{1}{2\pi} \frac{d\phi(t)}{dt}$$

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**Phase Modulation (PM):**

The instantaneous phase varies linearly with the message:

$$\phi_{PM}(t) = k_p \cdot m(t)$$

$$s_{PM}(t) = A_c \cos[\omega_c t + k_p m(t)]$$

For single-tone  $m(t) = A_m \cos \omega_m t$ :

$$s_{PM}(t) = A_c \cos[\omega_c t + k_p A_m \cos \omega_m t] = A_c \cos[\omega_c t + \beta_p \cos \omega_m t]$$

where  $\beta_p = k_p A_m$  = PM modulation index (rad).

Instantaneous frequency of PM:

$$f_i(t) = f_c - k_p A_m f_m \sin \omega_m t = f_c - \beta_p f_m \sin \omega_m t$$

Maximum frequency deviation:  $\Delta f_{PM} = k_p A_m f_m$  (increases with  $f_m$ )

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**Frequency Modulation (FM):**

The instantaneous frequency varies linearly with the message:

$$f_i(t) = f_c + k_f m(t)$$

Integrating to get phase:

$$\phi_{FM}(t) = 2\pi k_f \int_0^t m(\tau) d\tau$$

$$s_{FM}(t) = A_c \cos \left[ 2\pi f_c t + 2\pi k_f \int_0^t m(\tau) d\tau \right]$$

For single-tone  $m(t) = A_m \cos \omega_m t$ :

$$\phi_{FM}(t) = \frac{k_f A_m}{f_m} \sin \omega_m t = \beta_f \sin \omega_m t$$

$$s_{FM}(t) = A_c \cos[\omega_c t + \beta_f \sin \omega_m t]$$

where  $\beta_f = \frac{\Delta f}{f_m} = \frac{k_f A_m}{f_m}$  = FM modulation index.

Maximum frequency deviation:  $\Delta f = k_f A_m$  (independent of  $f_m$ )

### Bessel Function Expansion of FM:

Using the identity  $e^{j\beta \sin \theta} = \sum_{n=-\infty}^{\infty} J_n(\beta) e^{jn\theta}$ :

$$s_{FM}(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos(\omega_c + n\omega_m)t$$

The spectrum contains sidebands at  $f_c \pm n f_m$  with amplitudes  $A_c J_n(\beta)$ . Total power remains  $A_c^2/2$  regardless of  $\beta$ .

### Carson's Rule Bandwidth:

$$B_T = 2(\Delta f + f_m) = 2f_m(1 + \beta)$$

### FM vs PM Comparison:

| Property            | FM                     | PM                   |
|---------------------|------------------------|----------------------|
| $\phi(t)$           | $\propto \int m(t)dt$  | $\propto m(t)$       |
| $f_i(t)$            | $\propto m(t)$         | $\propto \dot{m}(t)$ |
| $\beta$ vs $A_m$    | Proportional           | Proportional         |
| $\beta$ vs $f_m$    | Inversely proportional | Independent          |
| $\Delta f$ vs $f_m$ | Independent            | Proportional         |

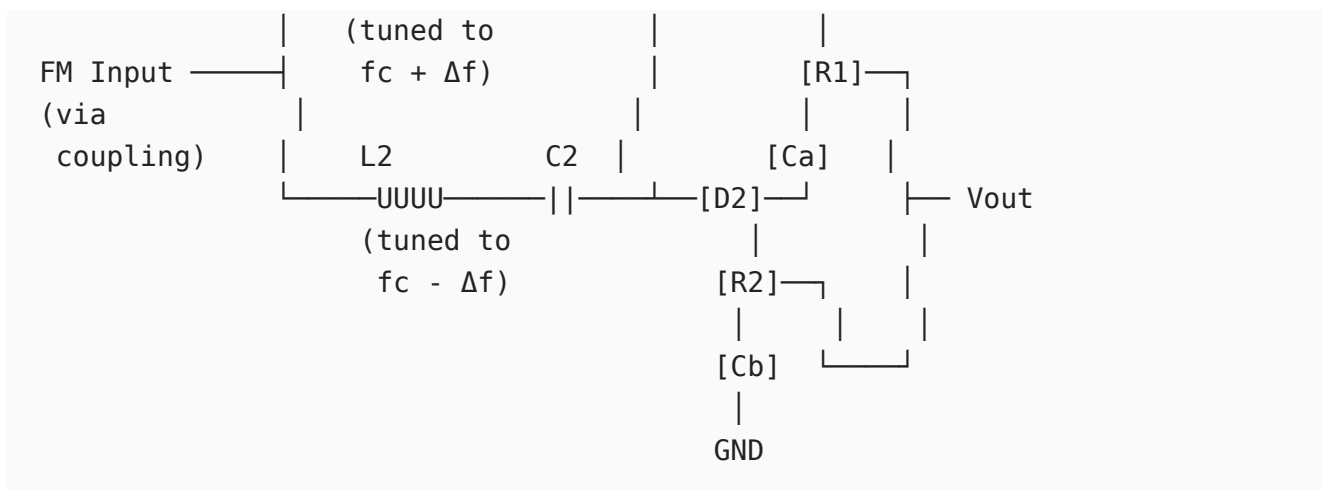
## ii. Illustrate the process of FM demodulation with balanced slope detector with the circuit diagram and characteristic curve.

### Principle:

The balanced slope detector (Travis discriminator) converts frequency variations to amplitude variations, which are then envelope-detected. Two tuned circuits detuned above and below  $f_c$  produce complementary amplitude responses, giving a linear output vs frequency characteristic.

### Circuit Diagram:





Two tuned circuits with opposite detuning. Diodes D1 and D2 are connected in opposing polarity.

### Operation Analysis:

At **carrier frequency**  $f = f_c$ :

- Both circuits equally detuned  $\rightarrow$  equal envelope outputs
- $V_{D1} = V_{D2} \rightarrow V_{out} = V_{D1} - V_{D2} = 0$

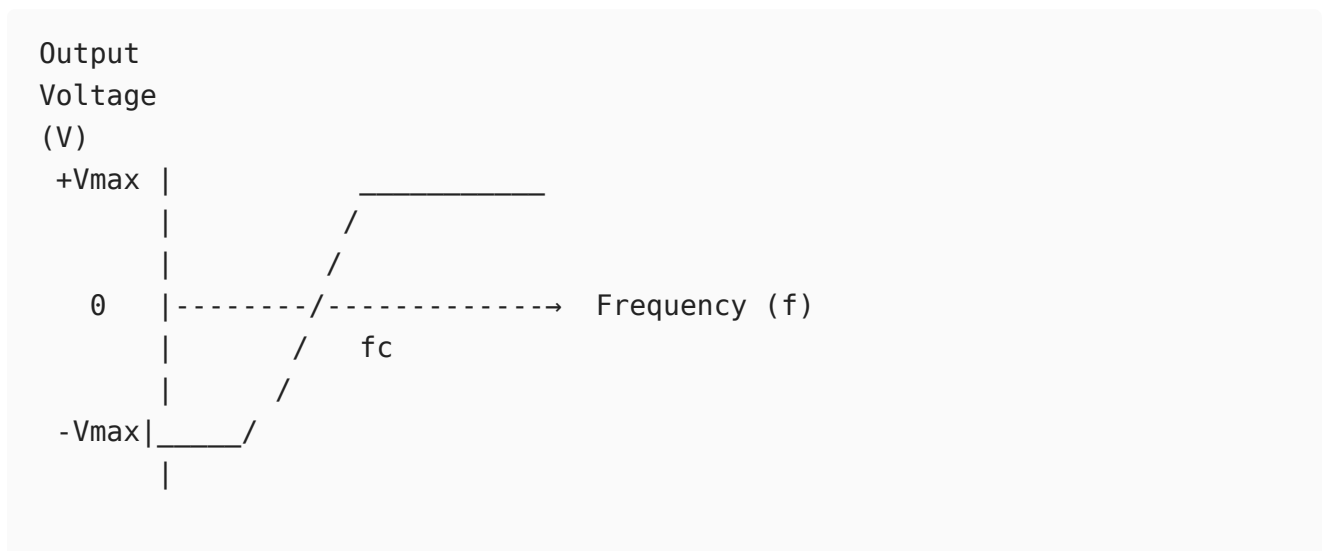
At  $f > f_c$  (positive deviation):

- L1C1 (tuned higher) moves toward resonance  $\rightarrow |V_{D1}|$  increases
- L2C2 (tuned lower) moves away from resonance  $\rightarrow |V_{D2}|$  decreases
- $V_{out} = V_{D1} - V_{D2} > 0 \rightarrow$  Positive output

At  $f < f_c$  (negative deviation):

- L1C1 moves further from resonance  $\rightarrow |V_{D1}|$  decreases
- L2C2 moves toward resonance  $\rightarrow |V_{D2}|$  increases
- $V_{out} = V_{D1} - V_{D2} < 0 \rightarrow$  Negative output

### Characteristic S-Curve:





← Linear region →  
( $f_c - \Delta f$ ) to ( $f_c + \Delta f$ )

The output is linear in the region  $f_c \pm \Delta f$ . Beyond this, the detector becomes nonlinear.

**Transfer characteristic** (linear region):

$$V_{out}(t) = k_d \cdot [f_i(t) - f_c] = k_d \cdot k_f \cdot m(t)$$

where  $k_d$  (V/Hz) is the detector sensitivity.

**Advantages:**

- Simple circuit, no phase-shift network required
- Good linearity within operating range

**Limitations:**

- Not self-limiting - sensitive to amplitude variations (AM noise)
  - Requires a limiter stage before it for proper FM demodulation
  - Bandwidth limited by the individual tuned circuit bandwidths
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## 12

**i. Illustrate the terms mean, correlation, covariance and ergodicity.**

**Mean (Statistical Average / Expected Value):**

For a random process  $X(t)$ , the mean at time  $t$  is:

$$\mu_X(t) = E[X(t)] = \int_{-\infty}^{\infty} x \cdot f_X(x; t) dx$$

It represents the ensemble average across all realizations at time  $t$ . Physically, it is the DC level of the random process. For a WSS process,  $\mu_X(t) = \mu_X = \text{constant}$ .

**Example:** For  $X(t) = A \cos(\omega_0 t + \Theta)$  with  $\Theta \sim U[0, 2\pi]$ :

$$\mu_X(t) = E[A \cos(\omega_0 t + \Theta)] = 0$$

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**Correlation:**

*Autocorrelation* measures similarity of a process with a time-shifted version of itself:

$$R_X(t_1, t_2) = E[X(t_1)X(t_2)]$$

For WSS:  $R_X(t_1, t_2) = R_X(\tau)$ ,  $\tau = t_1 - t_2$

$$R_X(\tau) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_X(x_1, x_2; \tau) dx_1 dx_2$$

$R_X(0) = E[X^2(t)]$  = total mean square power.

*Cross-correlation* between two processes:

$$R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)]$$


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### Covariance:

The autocovariance removes the mean before measuring correlation:

$$C_X(t_1, t_2) = E[(X(t_1) - \mu_X(t_1))(X(t_2) - \mu_X(t_2))]$$

$$\boxed{C_X(t_1, t_2) = R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2)}$$

For WSS with zero mean:  $C_X(\tau) = R_X(\tau)$

Variance (at  $t_1 = t_2$ ):

$$\sigma_X^2 = C_X(0) = R_X(0) - \mu_X^2 = E[X^2] - (E[X])^2$$

Two random variables are **uncorrelated** if  $C_{XY} = 0$ . Independent implies uncorrelated, but not vice versa (except for Gaussian).

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### Ergodicity:

A process is ergodic if **time averages equal ensemble averages** for any single realization.

*Mean-ergodic* condition:

$$\langle X(t) \rangle = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt = E[X(t)] = \mu_X$$

*Correlation-ergodic* condition:

$$\langle X(t)X(t + \tau) \rangle = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t)x(t + \tau) dt = R_X(\tau)$$

**Practical significance:** Ergodicity allows characterization of a random process from a single long observed waveform instead of requiring many ensemble samples. Most physical noise processes (e.g., thermal noise) are ergodic, which is why SNR can be measured from a single noise waveform trace. Ergodicity implies WSS, but WSS does not imply ergodicity.

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## ii. Interpret the process of autocorrelation and explain the properties of autocorrelation function.

### Autocorrelation Process:

The autocorrelation function (ACF)  $R_X(\tau)$  quantifies the degree of statistical dependence between the values of a random process at two time instants separated by  $\tau$ :

$$R_X(\tau) = E[X(t)X(t + \tau)]$$

Physically, it measures how quickly the process "remembers" or "forgets" its past values. It is computed by:

1. Taking one realization  $x(t)$
2. Creating a time-shifted copy  $x(t + \tau)$
3. Multiplying and taking the ensemble (or time) average

For a deterministic energy signal, the equivalent operation is the time-domain cross-correlation:

$$R_x(\tau) = \int_{-\infty}^{\infty} x(t)x(t + \tau) dt$$

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### Properties of the Autocorrelation Function:

#### Property 1 - Maximum at origin:

$$R_X(0) = E[X^2(t)] = \text{Total mean-square power}$$

$$|R_X(\tau)| \leq R_X(0) \quad \text{for all } \tau$$

$$\text{Proof: } E[(X(t) \pm X(t + \tau))^2] \geq 0 \Rightarrow R_X(0) \pm R_X(\tau) \geq 0$$

#### Property 2 - Even symmetry:

$$R_X(\tau) = R_X(-\tau)$$

The ACF is always an even function of  $\tau$ .

$$\text{Proof: } R_X(-\tau) = E[X(t)X(t - \tau)] = E[X(t + \tau)X(t)] = R_X(\tau)$$

#### Property 3 - Periodicity:

If  $X(t)$  contains a periodic component with period  $T_0$ , then  $R_X(\tau)$  is also periodic with period  $T_0$ :

$$R_X(\tau + T_0) = R_X(\tau)$$

#### Property 4 - DC component:

If  $E[X(t)] = \mu_X \neq 0$ , then:

$$\lim_{\tau \rightarrow \infty} R_X(\tau) = \mu_X^2$$

This represents the DC power. For a zero-mean process,  $R_X(\tau) \rightarrow 0$  as  $|\tau| \rightarrow \infty$ .

#### Property 5 - Wiener-Khinchin Theorem:

The ACF and the Power Spectral Density (PSD) form a Fourier transform pair:

$$S_X(f) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j2\pi f\tau} d\tau \quad \longleftrightarrow \quad R_X(\tau) = \int_{-\infty}^{\infty} S_X(f) e^{j2\pi f\tau} df$$

#### Property 6 - Non-negative PSD:

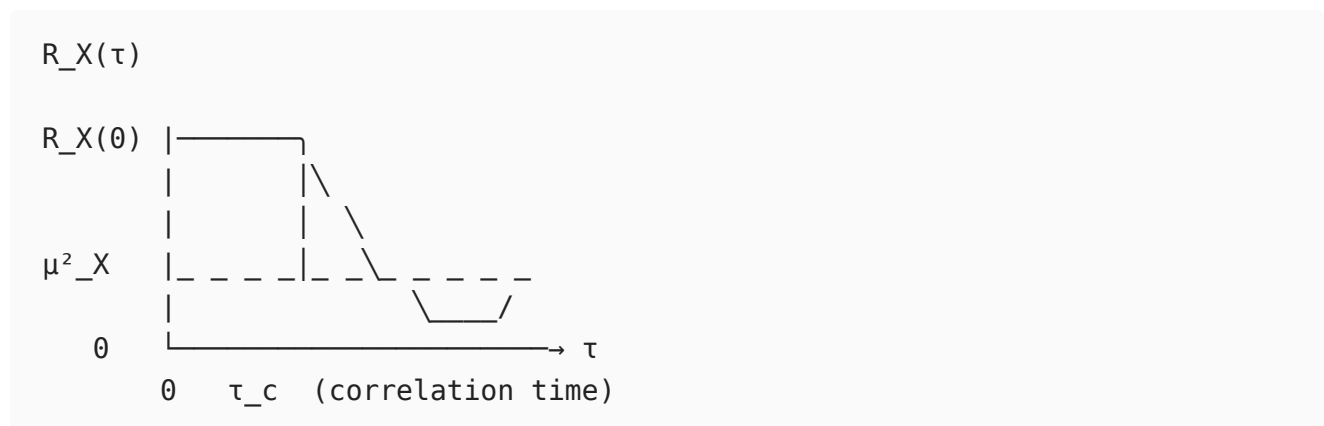
$$S_X(f) \geq 0 \quad \text{for all } f$$

This means the ACF must be a positive semi-definite function.

#### Property 7 - ACF at zero lag gives total power:

$$R_X(0) = \int_{-\infty}^{\infty} S_X(f) df = \text{Total power}$$

#### Graphical interpretation:



A narrow ACF (small  $\tau_c$ ) corresponds to a wideband PSD (fast-changing process). A wide ACF corresponds to a narrowband, slowly varying process.

## 13

**i. Apply the theory of internal and external noise sources to analyze their impact on communication systems.**

#### Classification of Noise:

Noise Sources  
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| External         |                   |              | Internal |      |         |
|------------------|-------------------|--------------|----------|------|---------|
| /                |                   | \            | /        |      | \       |
| Atmos.<br>static | Extra-<br>terres. | Man-<br>made | Thermal  | Shot | Flicker |

## External Noise Sources:

### 1. Atmospheric (Static) Noise:

Caused by lightning discharges and other natural electrical disturbances in the atmosphere.

- $\text{PSD} \propto 1/f^2$  - falls steeply with frequency
- Most severe below 30 MHz; negligible at microwave frequencies
- Limits AM broadcast receiver performance
- Equivalent noise temperature can be thousands of Kelvin at HF

### 2. Extraterrestrial Noise:

- *Solar noise*: Continuous radiation from the sun, intensifies during solar flares and sunspot activity. Significant at frequencies above 10 MHz.
- *Cosmic noise*: Background radiation from galactic sources (Milky Way). Significant from 8 MHz to 1.5 GHz. Characterized by a sky noise temperature  $T_{sky}$ .
- *Cosmic Microwave Background (CMB)*:  $T_{CMB} \approx 2.7$  K, relevant only for very sensitive radio telescopes.

### 3. Industrial / Man-Made Noise:

Generated by electrical equipment: motors, ignition systems, switching supplies, fluorescent lamps, transmission lines.

- Dominant in urban areas, particularly below 600 MHz
- Modeled as impulsive noise with spectral density  $\propto 1/f$

## Internal Noise Sources:

### 1. Thermal Noise (Johnson-Nyquist Noise):

Random motion of charge carriers due to thermal energy generates voltage fluctuations across any resistor:

$$\overline{v_n^2} = 4kTBR$$

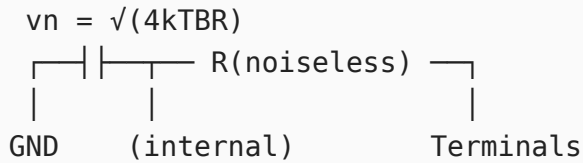
$$v_{n,rms} = \sqrt{4kTBR}$$

where  $k = 1.38 \times 10^{-23}$  J/K,  $T$  = absolute temperature (K),  $B$  = noise bandwidth (Hz),  $R$  = resistance ( $\Omega$ ).

Available noise power from a resistor:

$$P_n = kTB$$

Equivalent circuit: noise voltage source  $v_n$  in series with a noiseless resistor  $R$ .



PSD:  $S_v(f) = 2kTR$  (two-sided), or  $4kTR$  (one-sided) - flat (white noise).

## 2. Shot Noise:

Arises from the discrete nature of electron flow across a potential barrier (e.g., PN junction, vacuum tube):

$$\overline{i_n^2} = 2qI_{DC}B$$

where  $q = 1.6 \times 10^{-19}$  C,  $I_{DC}$  = average DC current. Also white noise. Significant in active devices at RF frequencies.

## 3. Flicker Noise (1/f noise):

Due to carrier trapping/recombination at semiconductor surface states:

$$S_n(f) \propto \frac{1}{f^\alpha}, \quad \alpha \approx 1$$

Dominant at low frequencies (below 1 kHz in bipolar transistors, up to hundreds of kHz in MOSFETs). Worsens oscillator phase noise near the carrier.

## 4. Transit-Time Noise:

At microwave frequencies where the signal period becomes comparable to electron transit time across a device. The electron stream induces random current pulses, increasing noise significantly.

## Impact on Cascaded System - Friis Formula:

For  $N$  cascaded stages with gains  $G_i$  and noise figures  $F_i$ :

$$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots + \frac{F_N - 1}{\prod_{i=1}^{N-1} G_i}$$

This shows that the first stage (LNA) dominates. A high-gain, low-noise LNA suppresses the noise contribution of all subsequent stages, which is the primary design criterion for RF receiver front-ends.

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## ii. Demonstrate how noise temperature is related to noise figure by applying SNR concepts.

### Noise Figure (F) - Definition from SNR:

Noise figure quantifies the degradation in signal-to-noise ratio introduced by a two-port network:

$$F = \frac{SNR_{in}}{SNR_{out}}$$

For an amplifier with power gain  $G$ , input signal power  $S_i$ , and input noise power  $N_i = kT_0B$  (standard reference temperature  $T_0 = 290$  K):

$$SNR_{in} = \frac{S_i}{kT_0B}$$
$$SNR_{out} = \frac{GS_i}{G \cdot kT_0B + N_{added,device}}$$

Therefore:

$$F = \frac{S_i/kT_0B}{GS_i/(GkT_0B + N_{device})} = 1 + \frac{N_{device}}{GkT_0B}$$

In dB:  $NF = 10 \log_{10}(F)$

The added noise from the device itself:  $N_{device} = (F - 1)GkT_0B$

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### Equivalent Noise Temperature ( $T_e$ ):

The equivalent noise temperature models the device's internal noise as if it were produced by a hypothetical noiseless device preceded by a resistor at temperature  $T_e$ :

$$T_e = (F - 1)T_0$$

Conversely:

$$F = 1 + \frac{T_e}{T_0}$$

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### Derivation:

Total output noise power of the amplifier:

$$\begin{aligned}N_o &= G \cdot N_i + G \cdot N_{internal} \\&= G \cdot kT_0B + G \cdot kT_eB \\&= GkB(T_0 + T_e)\end{aligned}$$

From the noise figure definition:

$$F = \frac{N_o}{GkT_0B} = \frac{GkB(T_0 + T_e)}{GkT_0B} = 1 + \frac{T_e}{T_0}$$

This confirms  $T_e = (F - 1)T_0$ .

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### SNR at output:

$$SNR_{out} = \frac{GS_i}{GkB(T_0 + T_e)} = \frac{S_i}{kB(T_0 + T_e)}$$

Or, using system noise temperature  $T_{sys} = T_0 + T_e = FT_0$ :

$$SNR_{out} = \frac{S_i}{kT_{sys}B}$$

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### Numerical relationship table:

| Noise Figure (dB) | F (linear) | $T_e$ (K)   |
|-------------------|------------|-------------|
| 0 dB              | 1.00       | 0 K (ideal) |
| 1 dB              | 1.26       | 75 K        |
| 3 dB              | 2.00       | 290 K       |
| 6 dB              | 4.00       | 870 K       |
| 10 dB             | 10.0       | 2610 K      |

---

### Receiver Sensitivity from SNR and Noise Figure:

Minimum detectable input signal:

$$S_{i,min} = F \cdot kT_0B \cdot SNR_{min}$$

In dBm (using  $kT_0 = -174$  dBm/Hz at 290 K):



$$S_{i,min}(\text{dBm}) = -174 + NF + 10 \log_{10}(B) + SNR_{min}(\text{dB})$$

**Example:**  $NF = 5 \text{ dB}$ ,  $B = 200 \text{ kHz}$ ,  $SNR_{min} = 12 \text{ dB}$ :

$$S_{i,min} = -174 + 5 + 53 + 12 = -104 \text{ dBm}$$

This directly shows why minimizing NF improves receiver sensitivity (lower detectable signal level).

---

## Set B

### Part-A

#### 1. Infer the need for modulation in communication systems.

- Antenna size becomes practical (e.g.,  $\lambda/4 = 75 \text{ cm}$  at  $100 \text{ MHz}$  vs.  $3.75 \text{ km}$  at  $20 \text{ kHz}$ )
  - Enables FDM - multiple users share one channel
  - FM provides significant SNR improvement ( $3\beta^2(\beta + 1)$  over baseband)
- 

#### 2. Write and explain the general expression for an AM signal.

$$s(t) = A_c[1 + \mu \cos \omega_m t] \cos \omega_c t$$

$A_c$  = carrier amplitude,  $\mu$  = modulation index ( $0 \leq \mu \leq 1$ ),  $\omega_m/\omega_c$  = message/carrier angular frequency. Contains carrier at  $f_c$  and sidebands at  $f_c \pm f_m$ .

---

#### 3. Identify whether distortion occurs if $\mu > 1$ .

Yes - **over-modulation**. The envelope  $[1 + \mu \cos \omega_m t]$  goes negative, causing an envelope detector to clip it, producing a distorted output that no longer resembles the original message.

---

#### 4. Using Carson's rule, calculate bandwidth if $\Delta f = 75 \text{ kHz}$ and $f_m = 15 \text{ kHz}$ .

$$B_T = 2(\Delta f + f_m) = 2(75 + 15) = \boxed{180 \text{ kHz}}$$


---

#### 5. Classify the FM signal as NBFM or WBFM if $\beta = 8$ .

**WBFM** - since  $\beta = 8 > 1$ , approximately  $\beta + 2 = 10$  significant sidebands per side;  
 $B_T = 2(\beta + 1)f_m = 18f_m$ . SNR improvement =  $3\beta^2(\beta + 1) \approx 32.4$  dB over baseband.

---

## 6. Summarize the important properties of ergodic and Gaussian random processes.

**Ergodic:** Time averages = ensemble averages; implies WSS; statistics measurable from a single realization.

**Gaussian:** Fully described by mean and ACF; linear transforms stay Gaussian; WSS  $\iff$  SSS; uncorrelated  $\Rightarrow$  independent.

---

## 7. Outline the essential properties that an autocorrelation function must satisfy.

- $R_X(0) = E[X^2(t)] \geq 0$  (total power)
  - $|R_X(\tau)| \leq R_X(0)$  (maximum at origin)
  - $R_X(\tau) = R_X(-\tau)$  (even symmetry)
  - $R_X(\tau) \rightarrow \mu_X^2$  as  $|\tau| \rightarrow \infty$
  - $\mathcal{F}\{R_X(\tau)\} = S_X(f) \geq 0$  (non-negative PSD)
- 

## 8. Apply the relationship between noise figure and noise temperature to analyze receiver characteristics.

$$T_e = (F - 1)T_0 \quad F = 1 + \frac{T_e}{T_0}, \quad T_0 = 290 \text{ K}$$

A 1 dB NF LNA gives  $T_e = 75$  K; minimum detectable signal  $S_{min} = FkT_0B \cdot SNR_{min}$ , so lower NF directly improves receiver sensitivity.

---

## 9. Demonstrate how thermal agitation in a resistor produces noise voltage and derive its mathematical expression.

Random electron motion (Brownian) in any resistor at  $T > 0$  K generates fluctuating voltage. By Nyquist's theorem:

$$\overline{v_n^2} = 4kTBR \implies v_{n,rms} = \sqrt{4kTBR}, \quad P_{available} = kTB$$

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## Part-B

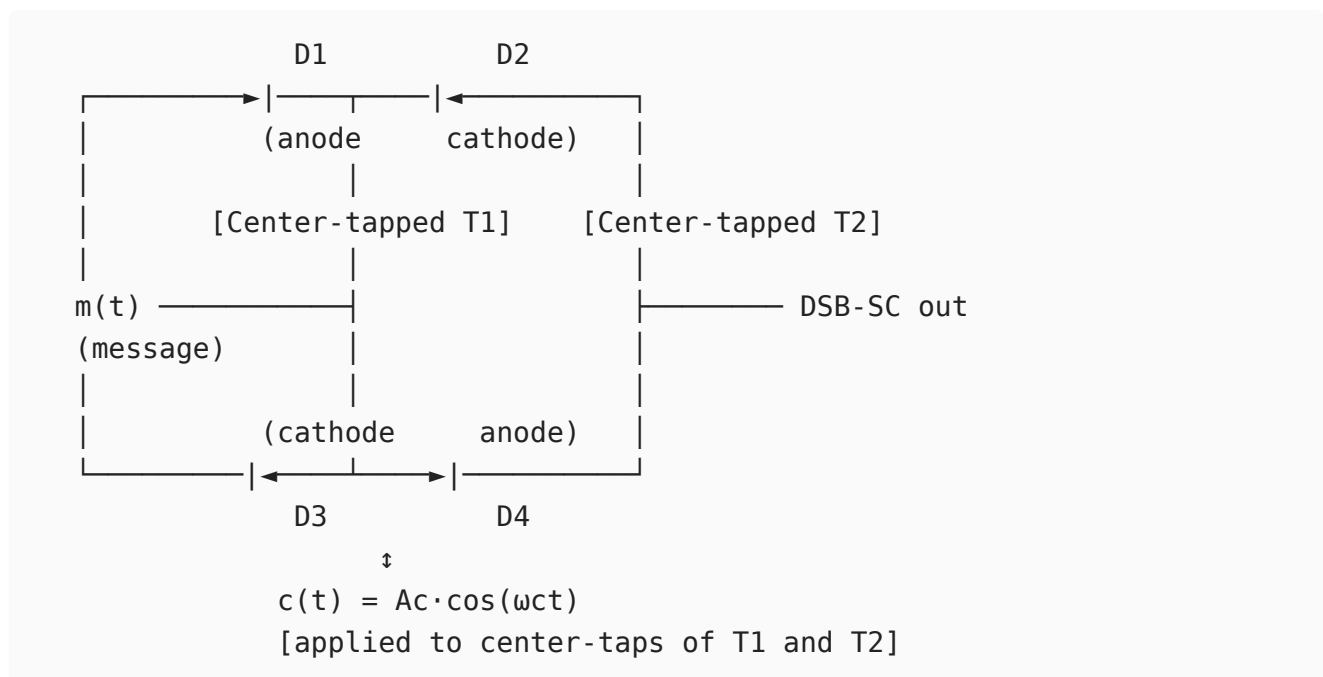
10

i. Construct the balanced modulator configuration for DSB-SC generation and describe its operation.

### Principle:

A balanced modulator suppresses the carrier component entirely, producing Double Sideband Suppressed Carrier (DSB-SC). This is achieved by using a balanced (symmetrical) circuit configuration where carrier components cancel by phase opposition.

### Ring (Lattice) Diode Modulator - Circuit:



### Operation - Positive half-cycle of $c(t)$ :

- D1 and D4 forward-biased, D2 and D3 reverse-biased
- Current path: top center-tap  $\rightarrow$  D1  $\rightarrow$  load  $\rightarrow$  D4  $\rightarrow$  bottom center-tap
- $m(t)$  is passed with **positive polarity** to the output transformer

### Operation - Negative half-cycle of $c(t)$ :

- D2 and D3 forward-biased, D1 and D4 reverse-biased
- $m(t)$  is passed with **inverted polarity** to the output transformer

Net effect:  $m(t)$  is multiplied by a square wave  $p(t)$  synchronized with the carrier.

### Mathematical Analysis:

The switching square wave:

$$p(t) = \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \cos[(2n+1)\omega_c t]$$

Output before filtering:

$$v_o(t) = m(t) \cdot p(t) = \frac{4}{\pi} m(t) \cos \omega_c t - \frac{4}{3\pi} m(t) \cos 3\omega_c t + \dots$$

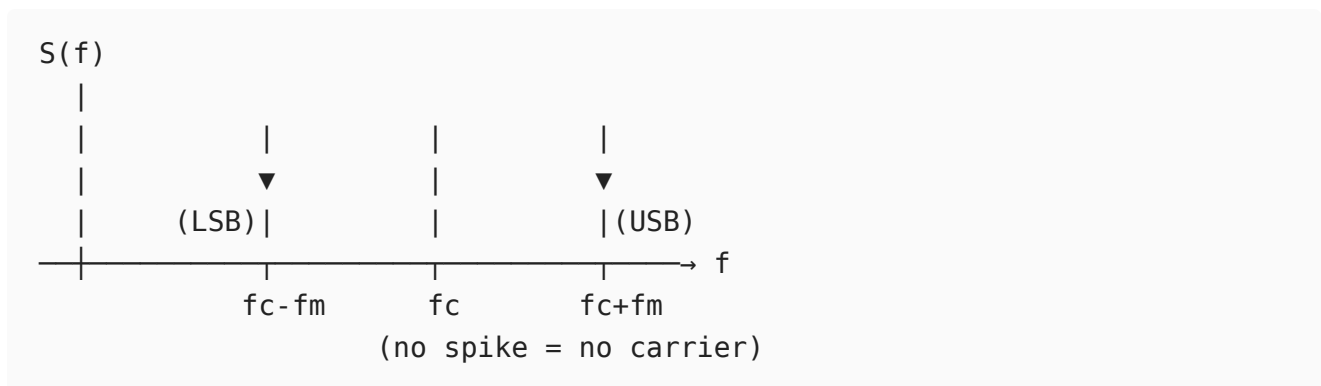
After BPF (passing only the term at  $\omega_c$ ):

$$s_{DSB-SC}(t) = \frac{4A_m}{\pi} \cos \omega_m t \cdot \cos \omega_c t$$

For  $m(t) = A_m \cos \omega_m t$ , expanding:

$$s_{DSB-SC}(t) = \frac{2A_m}{\pi} [\cos(\omega_c - \omega_m)t + \cos(\omega_c + \omega_m)t]$$

**Spectrum (no carrier component):**



**Carrier suppression** depends on the balance of the diode pairs. Practical circuits achieve 40-50 dB carrier suppression. Demodulation requires a coherent carrier reference at the receiver (the missing carrier must be regenerated via a pilot tone or Costas loop).

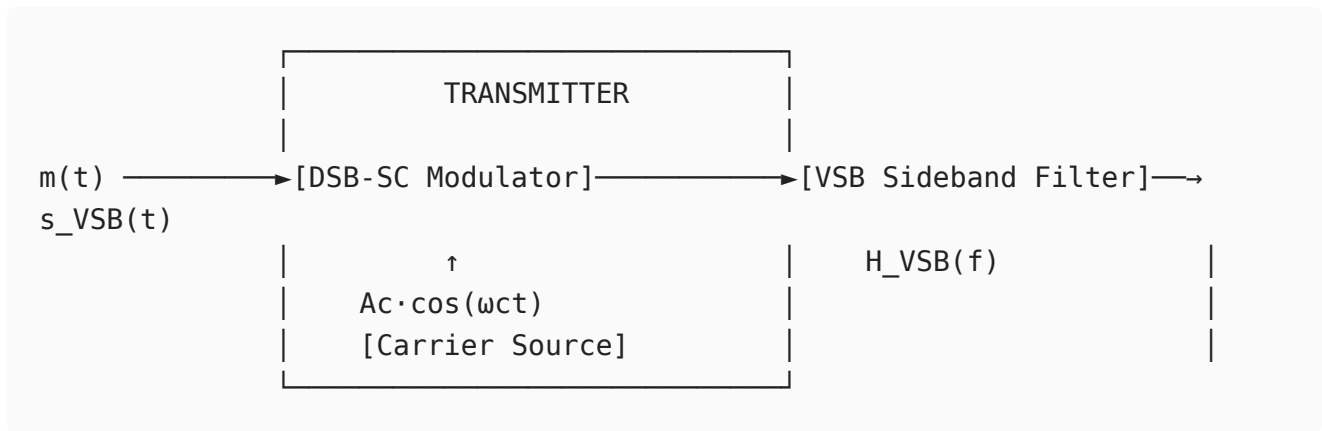
**Power efficiency:** 100% of transmitted power is in the sidebands (none wasted on carrier), making DSB-SC power-efficient compared to conventional AM (where carrier carries  $\sim 66\%$  of power at  $\mu = 1$ ).

**ii. Apply the concept of vestigial sideband modulation to design a transmitter and receiver block diagram and explain their operation.**

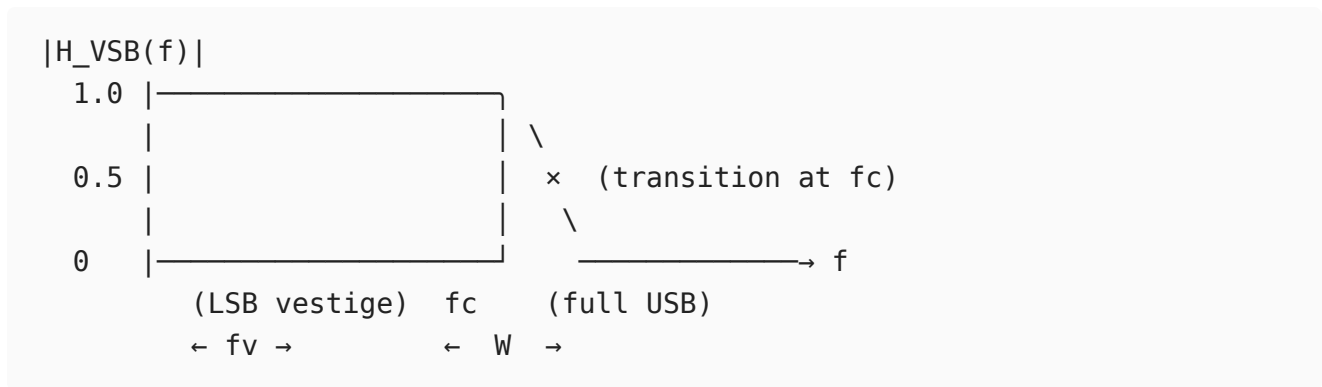
**Motivation for VSB:**

SSB requires an ideal sharp-cutoff filter at  $f_c$  (impractical near DC). DSB uses twice the necessary bandwidth. VSB is the engineering compromise: one full sideband is transmitted along with a controlled vestige of the other, using a gradual roll-off filter.

## VSF Transmitter Block Diagram:



## VSF Filter Response $H_{\text{VSB}}(f)$ :

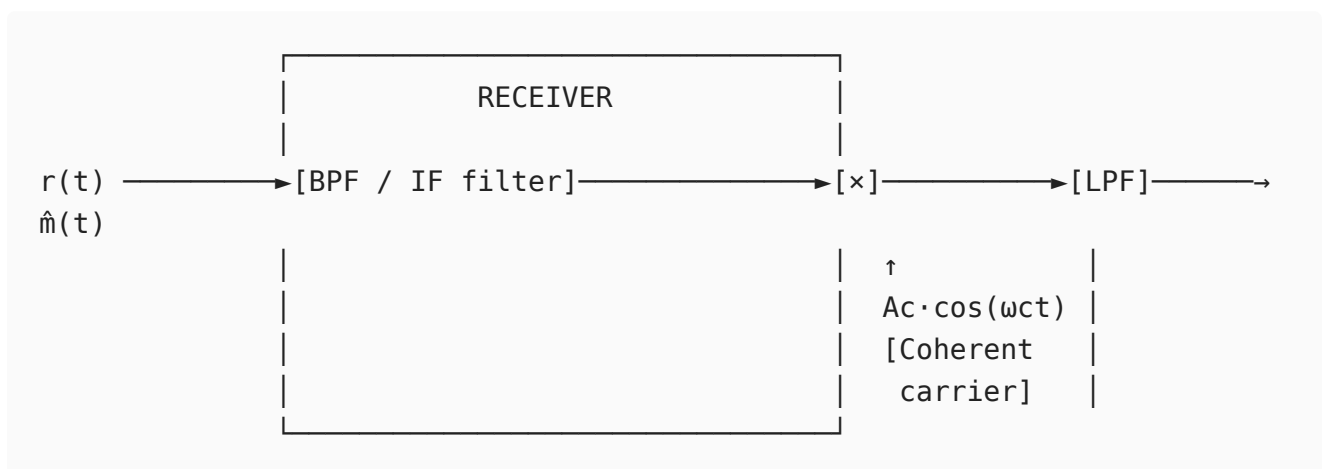


The filter satisfies the Nyquist vestigial symmetry condition:

$$H_{\text{VSB}}(f_c + f) + H_{\text{VSB}}(f_c - f) = 1, \quad |f| \leq W$$

This ensures that when demodulated, the two partial sideband contributions sum to the original full spectrum.

## VSF Receiver Block Diagram (Coherent Detection):



## Receiver Operation (coherent detection):

Received signal  $r(t) = s_{VSB}(t) + n(t)$ .

After BPF: Noise is limited to signal bandwidth.

After multiplying by coherent carrier  $\cos \omega_c t$  and applying LPF:

The vestigial symmetry condition ensures that the partial sidebands from USB and LSB vestige sum correctly:

$$\hat{m}(t) = \frac{1}{2}[m(t) * h_{VSB}(t)] + \frac{1}{2}[m(t) * h_{VSB}(t)] \propto m(t)$$

No additional equalization is needed if the VSB filter satisfies the symmetry criterion.

---

### Bandwidth comparison:

| Modulation | Bandwidth                      | Carrier transmitted |
|------------|--------------------------------|---------------------|
| DSB-AM     | $2W$                           | Yes                 |
| DSB-SC     | $2W$                           | No                  |
| SSB        | $W$                            | No                  |
| VSB        | $W + f_v$ (typ. $\sim 1.25W$ ) | Typically No        |

### Application - NTSC Television:

- Video baseband: DC to 4 MHz
- VSB filter retains: 4 MHz USB + 1.25 MHz vestige of LSB
- Total video bandwidth: 5.25 MHz within a 6 MHz channel
- Sound carrier: 4.5 MHz above video carrier (FM modulated)

**Advantage over SSB:** Near-DC video content (scene changes, slow pans) is preserved without the phase distortion that SSB would introduce near  $f = 0$ . The VSB filter's gradual roll-off is much easier to realize than an ideal brick-wall SSB filter.

---

## 11

**i. Show how Wideband FM is expressed mathematically and evaluate its performance characteristics relative to Narrowband FM.**

### Narrowband FM (NBFM) - $\beta < 0.3$ :

For small  $\beta$ , using the approximation  $\cos(\beta \sin \theta) \approx 1$  and  $\sin(\beta \sin \theta) \approx \beta \sin \theta$ :

$$s_{NBFM}(t) = A_c \cos \omega_c t - A_c \beta \sin \omega_m t \sin \omega_c t$$

Expanding using product-to-sum identities:

$$s_{NBFM}(t) \approx A_c \cos \omega_c t + \frac{A_c \beta}{2} [\cos(\omega_c + \omega_m)t - \cos(\omega_c - \omega_m)t]$$

Only three frequency components (carrier + 2 sidebands), bandwidth  $\approx 2f_m$ .

---

### Wideband FM (WBFM) - Mathematical Expression:

For arbitrary  $\beta$ , use the Bessel function expansion of the FM phasor  $e^{j\beta \sin \omega_m t}$ :

$$e^{j\beta \sin \omega_m t} = \sum_{n=-\infty}^{\infty} J_n(\beta) e^{jn\omega_m t}$$

Taking the real part:

$$s_{WBFM}(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos(\omega_c + n\omega_m)t$$

where  $J_n(\beta)$  are Bessel functions of the first kind, order  $n$ .

**Bessel coefficients**  $J_n(\beta)$  for key values:

| n | $\beta = 0.5$ (NBFM) | $\beta = 1$ | $\beta = 5$ (WBFM) |
|---|----------------------|-------------|--------------------|
| 0 | 0.938                | 0.765       | -0.178             |
| 1 | 0.242                | 0.440       | -0.328             |
| 2 | 0.031                | 0.115       | 0.047              |
| 3 | 0.003                | 0.020       | 0.365              |
| 4 | -                    | 0.002       | 0.391              |
| 5 | -                    | -           | 0.278              |

Significant sidebands (where  $|J_n(\beta)| > 0.01$ ) extend to approximately  $n_{max} \approx \beta + 2$ .

**Bandwidth (Carson's rule for WBFM):**

$$B_{WBFM} = 2(\Delta f + f_m) = 2f_m(1 + \beta)$$

**Total Power** (independent of  $\beta$ ):

$$P_{total} = \frac{A_c^2}{2} \sum_{n=-\infty}^{\infty} J_n^2(\beta) = \frac{A_c^2}{2}$$

---

## Performance Comparison - NBFM vs WBFM:

### SNR Analysis:

For FM, output SNR (post-detection) relative to input:

$$SNR_{FM,out} = \frac{3\beta^2(\beta + 1)A_c^2/2}{N_0W}$$

FM figure of merit compared to baseband:

$$\frac{SNR_{FM}}{SNR_{baseband}} = 3\beta^2(\beta + 1)$$

For NBFM ( $\beta = 0.5$ ): Improvement =  $3 \times 0.25 \times 1.5 = 1.125$  (only 0.5 dB above baseband)

For WBFM ( $\beta = 5$ ): Improvement =  $3 \times 25 \times 6 = 450$  (26.5 dB above baseband)

### Pre-emphasis and FM noise PSD:

FM demodulated noise has parabolic PSD:

$$S_{no}(f) = \frac{N_0 f^2}{A_c^2 k_f^2}, \quad |f| \leq W$$

This means high-frequency message components suffer more noise. Pre-emphasis (boost high frequencies before FM) and de-emphasis (attenuate at receiver) are used to equalize SNR across the audio band in commercial FM.

### Full Comparison Table:

| Parameter            | NBFM ( $\beta < 0.3$ )         | WBFM ( $\beta > 1$ )           |
|----------------------|--------------------------------|--------------------------------|
| Bandwidth            | $\approx 2f_m$                 | $2(\Delta f + f_m)$            |
| Spectrum             | 3 components                   | Many sidebands                 |
| SNR improvement      | $\approx 3\beta^2$ (small)     | $3\beta^2(\beta + 1)$ (large)  |
| Power in carrier     | $A_c^2 J_0^2(\beta)/2$ (high)  | Can be low (for high $\beta$ ) |
| Bandwidth efficiency | High                           | Low                            |
| Applications         | Two-way radio, 25 kHz channels | FM broadcast, satellite links  |
| Threshold effect     | Less critical                  | Requires input SNR > threshold |

**Threshold effect in WBFM:** Below a threshold CNR (typically 10 dB above  $kT_0B$ ), FM performance degrades rapidly ("FM clicks"). WBFM trades bandwidth for SNR improvement, valid only above threshold.

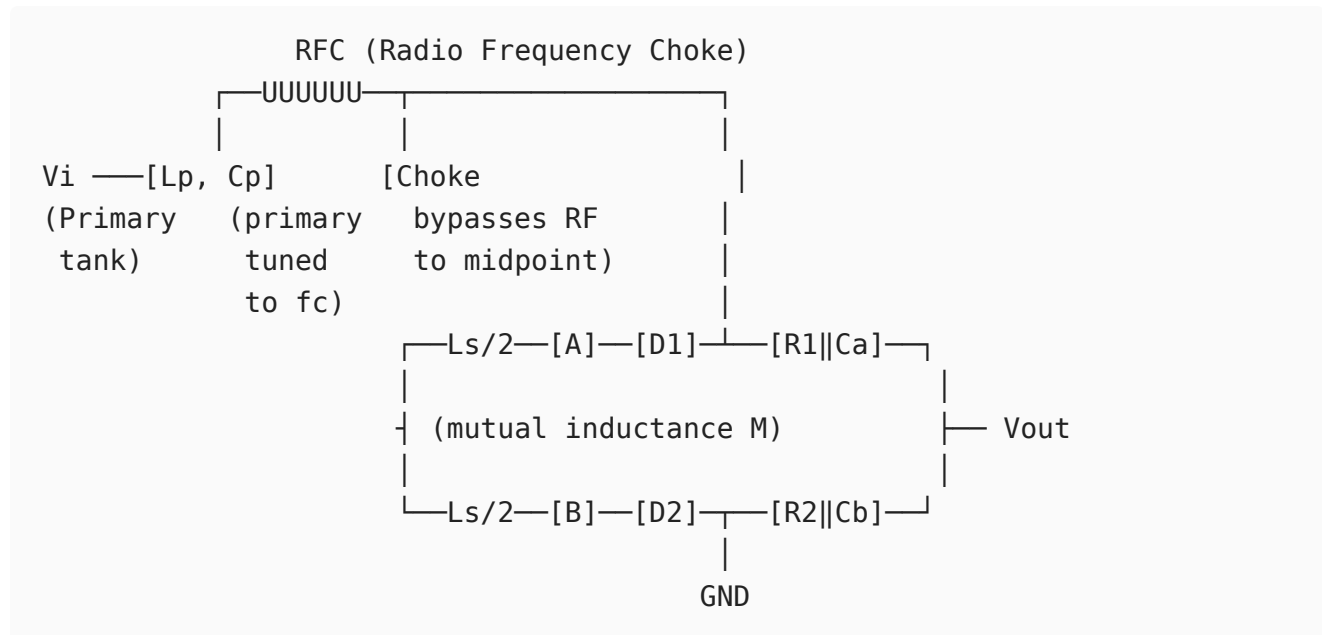
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## ii. Construct the phasor representation of voltages in a Foster-Seeley discriminator and explain how it recovers the modulating signal.

### Circuit Configuration:

The Foster-Seeley discriminator uses coupled resonant circuits and exploits the phase relationship between primary and secondary voltages to detect frequency deviation.



Key: The primary voltage  $V_1$  is fed to the center of the secondary via the RFC choke. Thus:

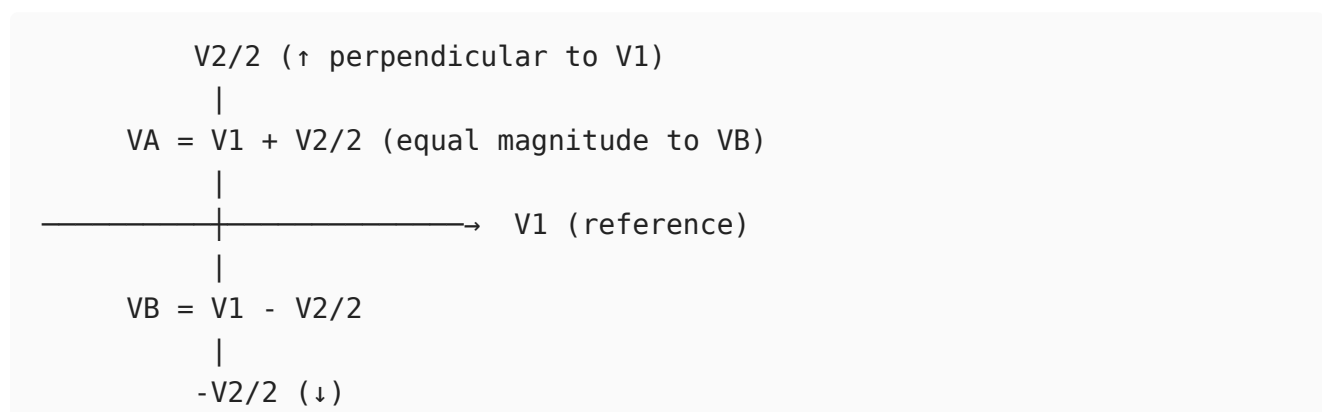
- Voltage at A:  $V_A = \frac{V_2}{2} + V_1$  (vectorially)
- Voltage at B:  $V_B = -\frac{V_2}{2} + V_1$  (vectorially)

where  $V_2$  is the secondary induced voltage.

### Phasor Analysis:

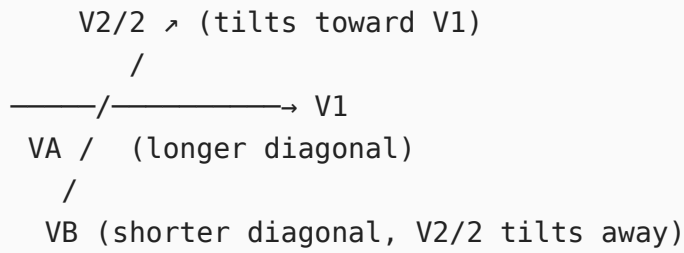
At resonance ( $f = f_c$ ):

- Mutual coupling voltage:  $V_2 \propto j\omega M I_1 \rightarrow$  secondary current  $I_2$  lags  $V_1$  by  $90^\circ$
- $V_2$  is perpendicular ( $90^\circ$ ) to  $V_1$



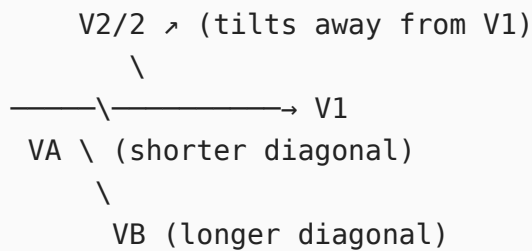
$$|V_A| = |V_B| \rightarrow V_{D1} = V_{D2} \rightarrow V_{out} = 0$$

Above resonance ( $f > f_c$ ), secondary becomes inductive, phase shift  $< 90^\circ$ :



$$|V_A| > |V_B| \rightarrow V_{out} > 0$$

Below resonance ( $f < f_c$ ), secondary becomes capacitive, phase shift  $> 90^\circ$ :



$$|V_A| < |V_B| \rightarrow V_{out} < 0$$

## Output Voltage:

$$\begin{aligned}
 V_{out} &= |V_A| - |V_B| \\
 &= \sqrt{V_1^2 + \left(\frac{V_2}{2}\right)^2 + V_1 \frac{V_2}{2} \cos \phi} - \sqrt{V_1^2 + \left(\frac{V_2}{2}\right)^2 - V_1 \frac{V_2}{2} \cos \phi}
 \end{aligned}$$

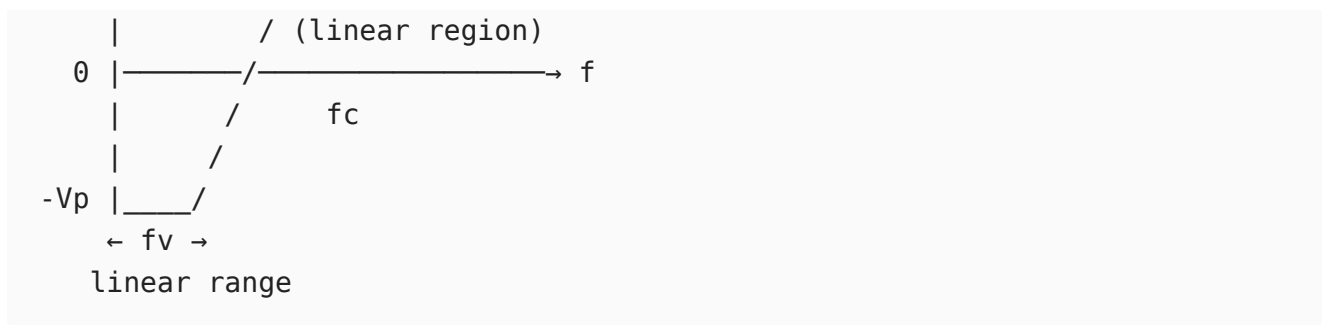
where  $\phi$  is the phase angle between  $V_1$  and  $V_2/2$ .

At  $f = f_c$ :  $\phi = 90^\circ$ ,  $\cos \phi = 0$ ,  $V_{out} = 0$

For small deviations:  $V_{out} \approx k_d(f - f_c) = k_d k_{fm}(t)$  (linear discriminator output)

## S-Curve Characteristic:





The slope of the linear portion ( $k_d$  in V/Hz) determines the discriminator gain. The linear range spans approximately  $\pm \Delta f_{max}$  around  $f_c$ .

### Signal recovery:

Since  $f_i(t) = f_c + k_f m(t)$ :

$$V_{out}(t) = k_d[f_i(t) - f_c] = k_d k_f m(t)$$

The original message  $m(t)$  is recovered with amplitude  $k_d k_f$ .

### Comparison with Balanced Slope Detector:

| Feature         | Balanced Slope                           | Foster-Seeley                 |
|-----------------|------------------------------------------|-------------------------------|
| Linearity       | Poor (uses two separate resonant slopes) | Better (uses phase variation) |
| Sensitivity     | Lower                                    | Higher                        |
| Bandwidth       | Narrower                                 | Wider                         |
| Limiter needed  | Yes                                      | Yes                           |
| Component count | More (two tuned circuits)                | Single coupled pair           |

Both require a limiter before them. The ratio detector is a modification of Foster-Seeley that provides inherent amplitude limiting.

## 12

### i. Formulate the mathematical representation of a random process using statistical parameters.

#### Definition:

A random process  $\{X(t, \zeta) : t \in T, \zeta \in S\}$  is a two-dimensional function:

- For fixed  $\zeta_k$ :  $X(t, \zeta_k) = x_k(t)$  is the  $k$ -th sample realization (deterministic waveform)
- For fixed  $t_k$ :  $X(t_k, \zeta)$  is a random variable

The ensemble is the collection of all realizations  $\{x_1(t), x_2(t), \dots, x_N(t)\}$ .

---

### First-Order Statistical Description:

First-order CDF:

$$F_X(x; t) = P[X(t) \leq x]$$

First-order PDF:

$$f_X(x; t) = \frac{\partial F_X(x; t)}{\partial x}$$

### Mean (First Moment):

$$\mu_X(t) = E[X(t)] = \int_{-\infty}^{\infty} x \cdot f_X(x; t) dx$$

### Mean-Square Value:

$$E[X^2(t)] = \int_{-\infty}^{\infty} x^2 f_X(x; t) dx$$

### Variance:

$$\sigma_X^2(t) = E[X^2(t)] - \mu_X^2(t) = E[(X(t) - \mu_X)^2]$$


---

### Second-Order Statistical Description:

Joint PDF at two time instants  $t_1, t_2$ :

$$f_X(x_1, x_2; t_1, t_2) = \frac{\partial^2 F_X(x_1, x_2; t_1, t_2)}{\partial x_1 \partial x_2}$$

### Autocorrelation Function:

$$R_X(t_1, t_2) = E[X(t_1)X(t_2)] = \iint x_1 x_2 \cdot f_X(x_1, x_2; t_1, t_2) dx_1 dx_2$$

### Autocovariance:

$$C_X(t_1, t_2) = R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2)$$

For WSS process ( $\tau = t_1 - t_2$ ):

$$R_X(\tau) = E[X(t)X(t + \tau)]$$

## Power Spectral Density (Wiener-Khinchin):

$$S_X(f) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j2\pi f\tau} d\tau$$

Total power:  $P_X = R_X(0) = \int_{-\infty}^{\infty} S_X(f) df$

---

## Example: Random Sinusoidal Process

$$X(t) = A \cos(\omega_0 t + \Theta), \quad \Theta \sim U[0, 2\pi]$$

**Mean:**

$$\mu_X(t) = E[A \cos(\omega_0 t + \Theta)] = \frac{A}{2\pi} \int_0^{2\pi} \cos(\omega_0 t + \theta) d\theta = 0$$

**Autocorrelation:**

$$\begin{aligned} R_X(t, t + \tau) &= E[A^2 \cos(\omega_0 t + \Theta) \cos(\omega_0 t + \omega_0 \tau + \Theta)] \\ &= \frac{A^2}{2} \cos \omega_0 \tau = R_X(\tau) \quad (\text{function of } \tau \text{ only}) \end{aligned}$$

This process is **WSS**: constant mean and  $\tau$ -dependent autocorrelation.

**PSD:**

$$S_X(f) = \mathcal{F} \left\{ \frac{A^2}{2} \cos \omega_0 \tau \right\} = \frac{A^2}{4} [\delta(f - f_0) + \delta(f + f_0)]$$

This shows discrete power at  $\pm f_0$ .

---

## Response of LTI System to a Random Process:

For WSS input  $X(t)$  through LTI system  $H(f)$ :

$$S_Y(f) = |H(f)|^2 S_X(f)$$

$$\mu_Y = \mu_X \cdot H(0)$$

$$R_Y(\tau) = \mathcal{F}^{-1} \{ |H(f)|^2 S_X(f) \}$$

This result (valid only for WSS input) is fundamental to noise analysis in communication receivers.

---

**ii. Show how the statistical properties determine whether a process is WSS or SSS and compare them.**

## Strict-Sense Stationary (SSS):

A process is SSS (or strongly stationary) if its complete joint statistical description is invariant to time translation. For any  $n$ , any set  $\{t_1, t_2, \dots, t_n\}$ , and any time shift  $\epsilon$ :

$$f_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = f_X(x_1, x_2, \dots, x_n; t_1 + \epsilon, t_2 + \epsilon, \dots, t_n + \epsilon)$$

This means **all statistical moments** (mean, variance, higher-order moments, and all joint distributions) are shift-invariant.

Consequences:

- Mean:  $E[X(t)] = \mu$  (constant)
- Autocorrelation:  $R_X(t_1, t_2) = R_X(\tau)$
- Third moment:  $E[X^3(t)] = \text{constant}$
- All higher moments: time-invariant

## Wide-Sense Stationary (WSS):

A weaker condition - only requires the first and second moments to be shift-invariant:

### WSS Conditions:

1.  $E[X(t)] = \mu_X$  (constant, independent of  $t$ )
2.  $R_X(t_1, t_2) = R_X(\tau)$ , where  $\tau = t_1 - t_2$

WSS does **not** require higher-order statistics to be stationary.

---

## Determination using Statistical Properties:

To check **WSS**:

1. Compute  $\mu_X(t)$ . If it varies with  $t$ , **not WSS**.
2. Compute  $R_X(t_1, t_2)$ . If it depends on both  $t_1$  and  $t_2$  (not just  $\tau = t_1 - t_2$ ), **not WSS**.

## Example test (WSS):

Process:  $X(t) = A(t) \cos(\omega_0 t)$  where  $A(t)$  has mean  $\mu_A$  and  $R_A(\tau)$ .

$$\mu_X(t) = \mu_A \cos \omega_0 t \quad \leftarrow \text{varies with } t: \text{ NOT WSS}$$

To check **SSS**: Must verify invariance of all joint PDFs under time shifts - practically intractable for most processes except Gaussian.

---

## Comparison Table:

| Aspect                  | WSS                                                                | SSS                                     |
|-------------------------|--------------------------------------------------------------------|-----------------------------------------|
| Conditions              | 2 conditions (mean + ACF)                                          | Infinite-order joint PDF invariance     |
| Strength                | Weak (necessary)                                                   | Strong (sufficient)                     |
| Practical verification  | Feasible                                                           | Usually intractable                     |
| Implication             | SSS $\Rightarrow$ WSS (always)                                     | WSS $\Rightarrow$ SSS only for Gaussian |
| All moments stationary? | No (only 1st and 2nd)                                              | Yes                                     |
| PSD valid?              | Yes (via Wiener-Khinchin)                                          | Yes                                     |
| Higher-order spectra    | Not guaranteed                                                     | Valid                                   |
| Example                 | WSS but not SSS: non-Gaussian process with stationary mean and ACF | White Gaussian Noise, thermal noise     |

---

### Key Result for Gaussian Processes:

For a **Gaussian** random process:

$$\text{WSS} \iff \text{SSS}$$

This equivalence holds because a Gaussian process is entirely characterized by its mean and covariance function. If these are shift-invariant (WSS conditions), then the entire probability structure (all joint PDFs) is automatically shift-invariant (SSS).

This equivalence is why Gaussian noise is so tractable - we only need to verify the two WSS conditions to conclude full stationarity.

---

### Impact on LTI System Analysis:

WSS is the minimum requirement for valid frequency-domain noise analysis:

$$S_Y(f) = |H(f)|^2 S_X(f)$$

This relation holds only if  $X(t)$  is WSS. SSS is stronger than needed - WSS suffices for all linear system noise performance calculations in communications.

## i. Show how noise figure influences overall receiver sensitivity and performance.

### Noise Figure and Sensitivity - Definitions:

Noise Figure:

$$F = \frac{SNR_{in}}{SNR_{out}} = \frac{S_i/N_i}{S_o/N_o} \geq 1$$

**Receiver Sensitivity** is the minimum input signal power  $S_{i,min}$  that produces a specified output  $SNR_{min}$ :

With input noise  $N_i = kT_0B$ :

$$S_{i,min} = F \cdot kT_0B \cdot SNR_{min}$$

In logarithmic form (using  $kT_0 \approx -174$  dBm/Hz at 290 K):

$$S_{i,min}(\text{dBm}) = -174 + NF(\text{dB}) + 10 \log_{10} B + SNR_{min}(\text{dB})$$

This equation directly shows that every 1 dB increase in noise figure raises the minimum detectable signal by 1 dB - degrading receiver sensitivity by 1 dB.

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### Cascaded Receiver Chain Analysis (Friis Formula):

For a practical receiver chain:

Antenna  $\rightarrow$  [LNA,  $G_1$ ,  $F_1$ ]  $\rightarrow$  [Mixer,  $G_2$ ,  $F_2$ ]  $\rightarrow$  [IF Amp,  $G_3$ ,  $F_3$ ]  $\rightarrow$  [Detector]

Total noise figure:

$$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots$$

**Critical observation:** If  $G_1 \gg 1$  (high gain LNA), the contributions from  $F_2, F_3, \dots$  become negligible. The LNA dominates system noise.

### Numerical example:

| Stage  | Gain               | NF                |
|--------|--------------------|-------------------|
| LNA    | 20 dB ( $G=100$ )  | 2 dB ( $F=1.58$ ) |
| Mixer  | -6 dB ( $G=0.25$ ) | 10 dB ( $F=10$ )  |
| IF Amp | 30 dB ( $G=1000$ ) | 6 dB ( $F=4$ )    |



$$F_{total} = 1.58 + \frac{10 - 1}{100} + \frac{4 - 1}{100 \times 0.25} = 1.58 + 0.09 + 0.12 = 1.79$$

$$NF_{total} = 10 \log(1.79) = 2.53 \text{ dB}$$

Despite the noisy mixer (10 dB NF), the high-gain LNA keeps total NF near 2.5 dB.

**Without LNA** (mixer first):

$$F_{total} = 10 + \frac{4 - 1}{0.25} = 10 + 12 = 22 \quad (13.4 \text{ dB})$$

This would degrade sensitivity by  $13.4 - 2.53 \approx 11 \text{ dB}$  - a huge degradation.

### Sensitivity floor calculations:

| Application   | NF (dB) | B (Hz)  | SNR_min (dB) | Sensitivity (dBm) |
|---------------|---------|---------|--------------|-------------------|
| FM radio      | 10      | 200 kHz | 12           | -99               |
| GSM cellular  | 8       | 200 kHz | 9            | -110              |
| GPS           | 2       | 2 MHz   | 14           | -135              |
| Satellite LNA | 0.5     | 36 MHz  | 10           | -108              |

### Noise Figure improvement techniques:

1. **LNA placement first:** Always place lowest NF stage at the antenna
2. **Minimize cable/filter loss** before LNA: Every 1 dB of loss = 1 dB added to NF
3. **Cryogenic cooling:** Reduces  $T_e$  (used in radio telescopes,  $T_e < 10 \text{ K}$ )
4. **GaAs HEMT LNAs:** Achieve NF of 0.3-0.5 dB at microwave frequencies

## ii. Demonstrate how narrowband noise is represented and analyze the behavior of its in-phase and quadrature components.

### Narrowband Noise - Definition:

Any broadband noise  $n(t)$  (e.g., white noise  $S_n(f) = N_0/2$ ) passed through a bandpass filter (BPF) of bandwidth  $B \ll f_c$  becomes narrowband noise.

### Canonical Representation:

Any narrowband noise process with center frequency  $f_c$  can be uniquely written as:

$$n(t) = n_I(t) \cos(2\pi f_c t) - n_Q(t) \sin(2\pi f_c t)$$

Or equivalently in envelope-phase form:

$$n(t) = r(t) \cos[2\pi f_c t + \psi(t)]$$

where:

- $n_I(t)$  = In-phase component (lowpass)
- $n_Q(t)$  = Quadrature component (lowpass)
- $r(t) = \sqrt{n_I^2(t) + n_Q^2(t)}$  = envelope (Rayleigh distributed if zero-mean Gaussian)
- $\psi(t) = \arctan\left(\frac{n_Q(t)}{n_I(t)}\right)$  = phase (uniform distributed)

Extraction of components:

$$n_I(t) = [n(t) \cos(2\pi f_c t)] * h_{LP}(t) \times 2$$

$$n_Q(t) = -[n(t) \sin(2\pi f_c t)] * h_{LP}(t) \times 2$$

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### Statistical Properties of $n_I(t)$ and $n_Q(t)$ :

For narrowband Gaussian noise with PSD  $S_n(f)$  and bandwidth  $B$ :

#### 1. Identical PSD:

$$S_{n_I}(f) = S_{n_Q}(f) = \begin{cases} S_n(f - f_c) + S_n(f + f_c), & |f| \leq B/2 \\ 0, & \text{otherwise} \end{cases}$$

For white noise through BPF:  $S_{n_I}(f) = S_{n_Q}(f) = N_0$  for  $|f| \leq B/2$ .

#### 2. Equal Power:

$$E[n_I^2(t)] = E[n_Q^2(t)] = E[n^2(t)] = N_0 B$$

The variance of each component equals the total noise power.

#### 3. Uncorrelated (and independent for Gaussian):

$$E[n_I(t)n_Q(t)] = 0$$

At the same time instant,  $n_I$  and  $n_Q$  are orthogonal. If  $n(t)$  is Gaussian, they are statistically independent.

#### 4. Both are lowpass with bandwidth $B/2$ .

#### 5. Gaussian: If $n(t)$ is Gaussian, $n_I$ and $n_Q$ are jointly Gaussian.

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## Behavior in AM Demodulation (Envelope Detector):

Received signal:  $r(t) = [A_c + m(t)] \cos \omega_c t + n(t)$

Expanding narrowband noise:

$$r(t) = [A_c + m(t) + n_I(t)] \cos \omega_c t - n_Q(t) \sin \omega_c t$$

Envelope:

$$R(t) = \sqrt{[A_c + m(t) + n_I(t)]^2 + n_Q^2(t)}$$

**High SNR** ( $A_c \gg n$ ):

$$R(t) \approx A_c + m(t) + n_I(t)$$

The quadrature component  $n_Q(t)$  is suppressed. Output noise is just  $n_I(t)$ , with power  $N_0 B$ .

$$SNR_{AM} = \frac{\mu^2 A_c^2 / 2}{N_0 W} \quad (\text{for tone modulation, } W = f_m)$$

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## Behavior in FM Demodulation:

The FM discriminator responds to instantaneous frequency. For received signal

$$r(t) = s_{FM}(t) + n(t):$$

Instantaneous phase of  $r(t)$  (high SNR approximation):

$$\phi_i(t) \approx \phi_{FM}(t) + \frac{n_Q(t)}{A_c}$$

Discriminator output (differentiates instantaneous phase):

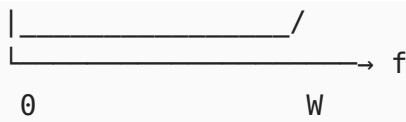
$$v_o(t) = \frac{1}{2\pi} \frac{d}{dt} \left[ \phi_{FM}(t) + \frac{n_Q(t)}{A_c} \right] = k_f m(t) + \frac{1}{2\pi A_c} \frac{dn_Q(t)}{dt}$$

FM output noise PSD (differentiation corresponds to  $\times j2\pi f$  in frequency domain):

$$S_{no,FM}(f) = \frac{(2\pi f)^2}{(2\pi A_c)^2} S_{n_Q}(f) = \frac{f^2 N_0}{A_c^2}$$

This **parabolic ( $f^2$ ) noise spectrum** is the defining characteristic of FM demodulated noise - high-frequency message components are noisier than low-frequency ones.





$$SNR_{FM} = \frac{3\beta^2(\beta + 1)A_c^2/2}{N_0W} = 3\beta^2(\beta + 1) \cdot SNR_{baseband}$$

**Rayleigh envelope distribution** (no signal):

$$f_r(r) = \frac{r}{\sigma^2} e^{-r^2/2\sigma^2}, \quad r \geq 0$$

where  $\sigma^2 = N_0B$  is the noise variance. This is the distribution of the noise envelope seen by an AM envelope detector in the absence of signal.